



Department of Computer Science and Business Systems Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: B.Tech, I & CSBS

Course Code & Title: CS3391 & Object Oriented Programming

Name of the Faculty member (s): Dr.S.Erana Veerappa Dinesh

Innovative Practice Description

- **Unit / Topic:** Unit I / Defining classes in Java

- **Course Outcome:** CO 1

- **Topic Learning Outcome:** TLO3

- **Activity Chosen:** Virtual Lab

- **Justification:**

The Virtual Laboratory is an interactive environment for creating and conducting simulated experiments: a playground for experimentation. A virtual laboratory is an on screen simulator that learners use to test ideas and observe results. Since the creation of classes and their objects is mandatory for the java programming language, hands on demonstration is given to the students through Virtual lab by using the following link

<https://java-iitd.vlabs.ac.in/>

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**

The course instructor explained the concept of java programming structures and creation of classes and objects in java. Figure 1 shows the Procedure for Creating Classes and their objects in java. He opened the Virtual Lab using aforementioned link and demonstrates the class creation and object creation in java to the students. Figure 2 and Figure 3 shows At last 10 minutes, the course instructor chosen few students and instructed them to execute and explain the examples in virtual lab by the students shown in the fig.4.

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PSO 1
CO 1	2	2	2	1

(1 – Low

2 – Moderate

3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 1	PO 2	PO 3	PSO 01
	2	2	1	2
Justification for correlation	Students could be able to apply basic syntax and principles of Java classes	Students could be able to have the hands-on with creating class and their objects on Virtual lab	Students could be able to recollect the basic principles of class creation in java	Students could be able to knowledge about the class creation and access the variable using object to their peer group of people.

- **Images / Screenshot of the practice:**



Fig 1: Procedure for Creating Class and their objects in java

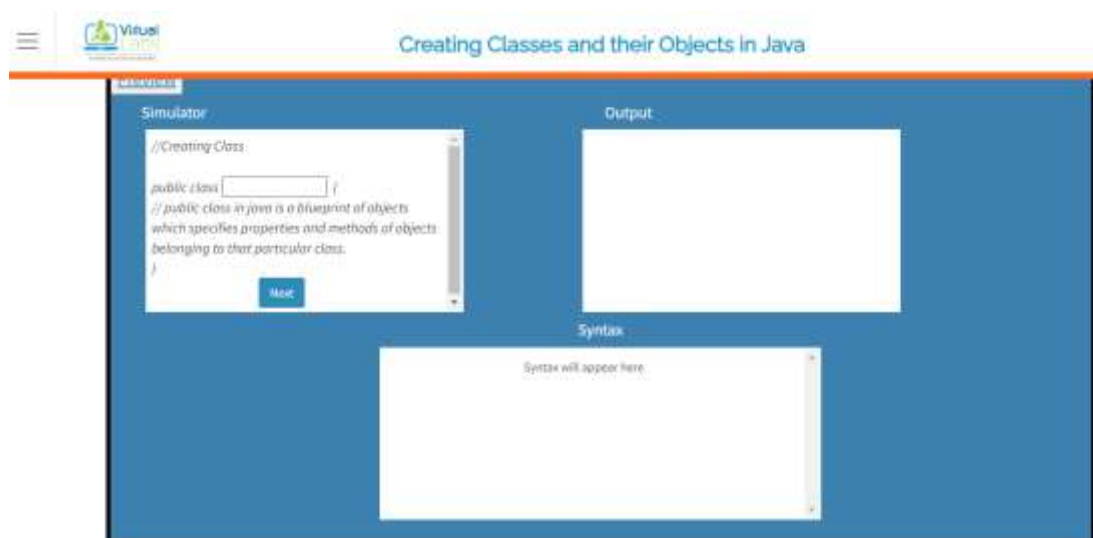


Fig:2 Demonstration on class creation

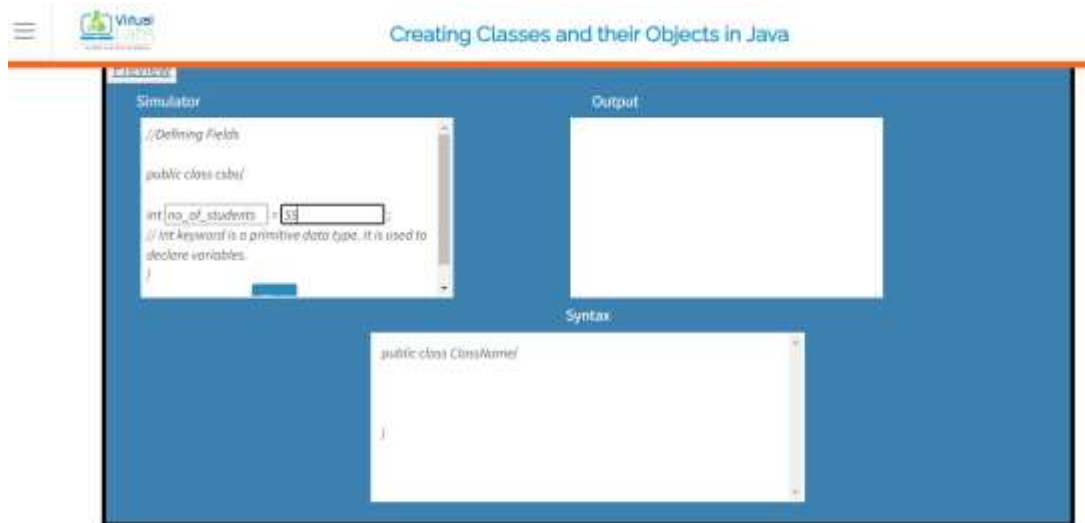


Fig:3 Demonstration on object creation



Fig:4 Demonstration by the student V.Akash

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Students easily understand the creation of classes, objects and its corresponding code.
 - Students said that the pretest and posttest is very useful to recollect the concepts.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students had understood the concepts of the classes and objects in java.
 - Most of the students were actively participated and enjoyed once they answered correctly in pretest.

- ❖ **Challenges faced in implementation:**

- Some students only involved in the demonstration activity.
 - The same activity will be planned in the laboratory session to make all the students familiar in the virtual environment.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/2021-2022/MGM_VII_Vlab_Demo.pdf
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/CS6801-MCAP-MGM.pdf>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/2021-2022/SM_I_Unit_3_Virtual_lab.pdf

Signature of Faculty Member**HOD**